APPSC INTEGRATED ENGINEERING INDEX & MICRO-TOPIC OVERLAP ANALYZER

27. ENGINEERING DRAWING & CAD

Underlined Highlights: **AEE PR/TW Extras** | **Draughtsman Gr-II Extras**

AEE PR/TW P-III: UNIT 1 (Drawing & CAD) DRAUGHTSMAN GR-II P-II: UNIT 1 (& U7 Geometry)

SUBJECT 27 • ENGINEERING DRAWING, PROJECTIONS & AUTOCAD

AEE PR/TW P-III UNIT 1 | DRAUGHTSMAN GR-II P-II UNIT 1

1. ENGINEERING DRAWING (AEE Notification 27/2026 — PR/Tribal Welfare Paper-III Unit 1) AEE PR/TW PAPER-III (U1)

Conic Sections including the **Rectangular Hyperbola. Cycloid, Epicycloid and Hypocycloid, Involute**. **Scales – Plain, Diagonal and Vernier Scales**. **Principles of Orthographic Projections – Conventions – Projections of Points and Lines – Projections of Plane regular geometric figures, Projections of Regular Solids – Auxiliary Views. Sections or Sectional views of Right Regular Solids – Prism, Cylinder, Pyramid, Cone – Auxiliary views – Sections of Sphere. Development of Surfaces of Right Regular Solids – Prism, Cylinder, Pyramid and Cone. Principles of Isometric Projection – Isometric Scale – Isometric Views – Conventions – Isometric Views of Lines, Plane Figures, Simple and Compound Solids – Isometric Projection of objects having non-isometric lines. Isometric Projection of Spherical Parts. Conversion of Isometric Views to Orthographic Views and Vice-versa – Conventions – Auto-CAD: Basic principles**.

1. Drawing (APPSC Draughtsman Grade-II Paper-II Unit 1 + Unit 7h,i Geometry) DRAUGHTSMAN GR-II P-II (U1)

a) Importance of B.I.S. – introduction of Code of Practice for Architectural and Building Drawings (IS: 962-1989). b) Layout of drawing. Lines, Lettering, Dimensioning, **Scales and Projection**. *(Plus Unit 7h,i: Geometrical properties of lines, angles, triangles, regular polygons, circles, parallelogram, **parabola and ellipse**).*

100% Common Core (AEE PR + Draughtsman Gr-II)	Extra Depth ONLY in AEE PR/TW Paper-III (Unit 1)	Extra Standards ONLY in Draughtsman Gr-II (Unit 1)
• Drawing Conventions (Lines, Lettering, Dimensioning) • Representative Fraction (RF), Plain & Diagonal Scales • Conic Sections: Ellipse, Parabola, Hyperbola • Orthographic Projections (1st & 3rd Angle)	• **Vernier Scales (Least Count)** • **Rectangular Hyperbola, Cycloid, Epicycloid, Hypocycloid & Involute** • **Projections of Points, Lines, Planes, and Solids** • **Auxiliary Views, Section of Solids & Development of Surfaces** • **Isometric Scale & Isometric Views**	• **IS: 962-1989 (Architectural and Building Drawing Code)** • **Drawing Sheet Sizes (A0 to A4), Title Block dimensions & Folding** • **Chord of Scales & Comparative Scales** • **AutoCAD commands (Layers, Blocks, Hatching, Shortcuts)**

SUBJECT 25A • SURVEYING FUNDAMENTALS, CHAIN/COMPASS, LEVELLING & CONTOURING

AEE CIVIL P-III UNIT 9 (PART 1) | DRAUGHTSMAN GR-II P-II UNITS 2 & 3

9. SURVEYING AND TRANSPORT ENGINEERING (Part 1: Fundamental Surveying & Levelling) — AEE CIVIL PAPER-III (U9)

Classification of surveys, **Scales**, Accuracy, Measurement of distances, Direct and indirect methods, **Optical and electronic devices**, Measurement of directions, Prismatic compass, **Local attraction**, Measurement of elevations, Spirit and **trigonometric leveling**, Contours, **Digital Elevation Modeling (DEM) concept**, **Map preparation by plane tabling**, **Map substitutes**.

2. Surveying & 3. Levelling (APPSC Draughtsman Grade-II Paper-II Units 2 & 3) — DRAUGHTSMAN GR-II P-II (U2, U3)

2. Surveying: a–h) Introduction, principles, classification, chaining, bearing, meridian, **Planimeter and pantograph**.
3. Levelling: a–i) **Auto level introduction, principle, graduation & types of staffs, temporary and permanent adjustments, reduction of levels (HI & Rise-and-Fall)**. j–l) Contouring: Definition, methods, **Interpolation, contour gradient**.

100% Common Core (AEE Civil U9 + Draughtsman U2, U3)	Extra Sub-Topics ONLY in AEE Civil Paper-III (U9)	Extra Sub-Topics ONLY in Draughtsman Gr-II (U2, U3)
• Principles of Surveying (Whole to part) • Plane vs Geodetic Surveying • Chain & Tape Corrections • Compass Surveying & Local Attraction • Spirit Leveling & Contouring Methods	• **Tacheometry (Stadia & Tangential systems)** • **Trigonometric Leveling** • **Plane Table Resection (Bessel's/Lehmann's)** • **Digital Elevation Model (DEM) & Orthophoto**	• **Planimeter & Pantograph / Eidograph** • **Auto-Level internal mechanism** • **Levelling Staff types & Two-peg test**

SUBJECT 25B • THEODOLITE, TRAVERSING, TOTAL STATION, GPS & PHOTOGRAMMETRY

AEE CIVIL P-III UNIT 9 (PART 2) | DRAUGHTSMAN GR-II P-II UNITS 4, 5 & 6

9. SURVEYING AND TRANSPORT ENGINEERING (Part 2: Theodolite, GPS & RS) — AEE CIVIL PAPER-III (U9)

Theodolites, Types, Establishment of control by triangulations and traversing, Measurement and adjustment of observations (Least Squares), Computation of coordinates, Field astronomy, GPS concept, Photogrammetry, Remote sensing.

4. Theodolite, 5. Total Station & 6. GPS (Draughtsman Grade-II P-II Units 4–6) — DRAUGHTSMAN GR-II P-II (U4–U6)

4. Theodolite: Measurement of angles, Gale's Traverse Table, closing errors. **5. Total Station:** EDM principle, coordinate systems. **6. GPS & RS:** WGS-84, C/A & P-code, GPS biases, satellite segments.

100% Common Core (AEE Civil U9 + Draughtsman U4–U6)	Extra Sub-Topics ONLY in AEE Civil Paper-III (U9)	Extra Sub-Topics ONLY in Draughtsman Gr-II (U4–U6)
• Theodolite Fundamental Axes & Angles • Latitude & Departure, Closing Error & Bowditch Rule • Triangulation & Photogrammetry basics	• Theory of Errors & Least Squares Adjustment • Field Astronomy (Celestial sphere, Azimuth)	• Gale's Traverse Table computations • Total Station & EDM Physics • Deep GPS Architecture (WGS-84, L1/L2, C/A & P-codes)

SUBJECT 28 • PRINCIPLES OF ELECTRICAL & ELECTRONIC ENGINEERING

AEE PR/TW PAPER-III UNIT 9 (100% EXCLUSIVE TO AEE PR/TW)

9. Principles of Electrical and Electronic Engineering (PR/TW Paper-III Unit 9) AEE PR/TW PAPER-III (U9)

i. Electrical Circuits: R-L-C Parameters, KCL/KVL, Star-Delta transformation.
ii. Single Phase AC Circuits: RMS, Average, Form Factor, Impedance, Admittance, Power Factor.
iii. P-N Junction Diode: Diode equation, V-I characteristics, Capacitances.
iv. Bipolar Junction Transistor (BJT): CB, CE, CC configurations, Feedback amplifiers.
v. Special Purpose Devices: Zener diode characteristics and voltage regulation.
vi. Rectifiers and Linear ICs: Half-wave, Full-wave & Bridge Rectifiers, Op-Amp applications.

PR/TW Exclusive Module	High-Yield Formula & Concept Checklist for Civil/Mech Aspirants
1. DC Networks & AC Circuits	• Passive R, L, C parameters; Source transformation; KCL & KVL; Star-Delta conversion ($R_\Delta = 3 R_Y$) • RMS Value ($I_m/\sqrt{2}$), Average Value ($2I_m/\pi$), Form Factor (1.11), Impedance Z, Admittance Y = 1/Z, Power Factor cos φ
2. Electronics & Op-Amps	• Shockley diode equation; BJT configurations (CE, CB, CC); Zener diode voltage regulator • Rectifiers (Half-wave, Center-tapped, Bridge — Ripple factor & PIV); Op-Amp (Inverting & Non-inverting gain)

SUBJECT 29 • COMPUTER FUNDAMENTALS, MEMORY HIERARCHY & OS FILE OPERATIONS

AEE PR/TW PAPER-III UNIT 10 (100% EXCLUSIVE TO AEE PR/TW)

10. COMPUTER FUNDAMENTALS (PR/Tribal Welfare Paper-III Unit 10) AEE PR/TW PAPER-III (U10)

Basics of computer and its operation: Functional Components and their inter-connections, concept of Booting, Use of Operating System for directory listing, hierarchical directory structure, renaming, deleting files/folders, Memory and types of memory, working of USB flash drives, copying files, concepts of path and pathname, switching between tasks, installation/removal of applications .

Unit 10 Micro-Topic	Exact Technical Concepts & Commands Tested in AEE PR/TW
1. Architecture & Booting	• Von Neumann Architecture (ALU, CU, Registers); System Buses (Data, Address, Control) • Booting Sequence (POST -> BIOS/UEFI -> MBR -> OS Kernel); Memory Hierarchy (Registers -> Cache -> RAM -> Storage)
2. OS File Management	• Hierarchical tree directory structure; Absolute vs Relative Pathnames • CLI & GUI Commands (`dir`/`ls`, `md`/`mkdir`, `ren`/`mv`, `del`/`rm`, `copy`/`cp`), Task switching & App installation

SUBJECT 31 • UNITS, ARITHMETIC, ALGEBRA, GEOMETRY, MENSURATION & SURVEY RULES

DRAUGHTSMAN GR-II P-II UNIT 7 | OVERLAPS WITH AEE CIVIL SURVEYING

7. Units, arithmetic, algebra, geometry, mensuration (Draughtsman Grade-II P-II Unit 7) DRAUGHTSMAN GR-II P-II (U7)

a–g) Units, Arithmetic & Algebra: MKS, FPS & CGS conversions, LCM, Fractions, Decimals, Linear & Quadratic equations.
h–n) Geometry, Mensuration & Trig: Properties of polygons, circles, cylinders, spheres, cones, Heights & Distances.
o–r) Surveying Area & Volume Rules: Trapezoidal rule, Simpson's 1/3 rule, Prismoidal rule.

100% Common Core (Draughtsman U7 + AEE Civil U9)	Formulae to Memorize for Irregular Area & Earthwork Volumes
• **Surveying Earthwork Rules:** Shared between Draughtsman Unit 7 and AEE Civil Surveying! • **2D/3D Mensuration:** Shared with Group-1/Group-2 Aptitude.	• **Trapezoidal Rule:** $\Delta = \frac{d}{2} \left[(O_1 + O_n) + 2(O_2 + O_3 + \dots)\right]$ • **Simpson's 1/3 Rule (Even intervals / Odd ordinates):** $\Delta = \frac{d}{3} \left[(O_1 + O_n) + 4(\sum \text{Even}) + 2(\sum \text{Odd})\right]$ • **Prismoidal Formula (Volumes):** $V = \frac{d}{3} \left[A_1 + A_n + 4(\sum \text{Even}) + 2(\sum \text{Odd})\right]$